The Rank of the Symmetrizing Product: Injectivity, Cascade Surjectivity, and the Formula $n!/2^{\lfloor n/2 \rfloor}$

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Abstract

We study the staircase symmetrizing product $\Pi = \prod(1+s_{i_j})/2$ for the longest element $w_0 \in \mathfrak{S}_n$, acting on the left regular representation. We prove an *Injectivity Lemma* showing rank drops occur only at the P_1 steps, a *First Block Surjectivity* theorem identifying the image after the first pass as $+1(s_{n-1})$, a *Cascade Surjectivity* lemma showing how eigenspace conditions propagate through the cascade, and a *Trace Preservation* lemma controlling s_1-traces. These combine to prove the first two halvings unconditionally and yield the rank formula $\mathrm{rank}(\Pi) = n!/2^{\lfloor n/2 \rfloor}$ modulo a halving lemma verified for $n \leq 8$.

1 Setup

Let $\mathfrak{S}_n$ act on $\{1, \ldots, n\}$ with simple transpositions $s_i = (i, i{+}1)$. In the left regular representation, s_i acts by $s_i \cdot e_w = e_{s_i w}$. Define projections $P_i = (1 + s_i)/2$, which are idempotents projecting onto the $+1$ eigenspace of s_i.

Definition 1 (Staircase product)**.** The staircase reduced word for $w_0 \in \mathfrak{S}_n$ consists of $n{-}1$ *passes*: pass j (for $j = 1, \ldots, n{-}1$) applies the projections $P_1, P_2, \ldots, P_{n-j}$ in sequence. The full product is

$$\Pi = \underbrace{P_1}_{\text{pass } n-1} \cdot \underbrace{P_2 P_1}_{\text{pass } n-2} \cdots \underbrace{P_{n-1} \cdots P_2 P_1}_{\text{pass } 1}.$$

The rightmost pass is applied first.

Notation. Write $+1(s_k)$ for the $+1$ eigenspace of s_k and $-1(s_k)$ for the -1 eigenspace. Write W_j for the image of $\mathbb{C}[\mathfrak{S}_n]$ after the first j passes have been applied.

2 The Injectivity Lemma

Lemma 2 (Triple Lemma)**.** *In $\mathbb{C}[\mathfrak{S}_n]$ with $n \geq 3$, for any $1 \leq k \leq n-2$:*

$$\ker(P_{k+1}) \cap \mathrm{im}(P_k) = \{0\}.$$

Equivalently, $+1(s_k) \cap -1(s_{k+1}) = \{0\}$.

Proof. By symmetry, it suffices to prove the case $k = 1$. Write $v = \sum c_w e_w$ and group terms by the triple of values $\{a, b, c\}$ occupying positions $1, 2, 3$ and by the tail $w|_{\{4,\ldots,n\}}$. Within each 6-element group, the condition $s_1 v = v$ forces equal coefficients on each s_1-orbit, leaving three variables α, β, γ. The condition $s_2 v = -v$ gives $\beta = -\alpha$, $\gamma = -\alpha$, and $\gamma = -\beta$. Together: $\gamma = -\gamma$, so $\alpha = \beta = \gamma = 0$. $\square$

Theorem 3 (Within-Pass Injectivity)**.** *In the sequential application $P_1, P_2, \ldots, P_m$ (with $m \leq n-1$), P_j is injective on $P_{j-1}(\cdots P_1(W)\cdots)$ for all $j \geq 2$ and any initial subspace W.*

Proof. At step $j \geq 2$, the image of P_{j-1} lies in $\mathrm{im}(P_{j-1}) = +1(s_{j-1})$. The kernel of P_j intersects $+1(s_{j-1})$ trivially by Lemma 2. □

Corollary 4. *In the staircase product Π, rank drops occur only at the P_1 steps. The total rank is $n!/2^d$ where d is the number of passes whose initial P_1 step causes a nontrivial rank reduction.*

3 First Block Surjectivity

Theorem 5 (First Block Surjectivity)**.** *After pass 1 (applying $P_1, P_2, \ldots, P_{n-1}$ to $\mathbb{C}[\mathfrak{S}_n]$), the image is exactly $+1(s_{n-1})$:*

$$W_1 = P_{n-1} \cdots P_2 P_1(\mathbb{C}[\mathfrak{S}_n]) = +1(s_{n-1}).$$

In particular, $\dim(W_1) = n!/2$.

Proof. After P_1: the image is $+1(s_1)$, which has dimension $n!/2$. After P_2: the image lies in $+1(s_2)$. By Theorem 4, P_2 is injective on $+1(s_1)$, so $\dim(P_2(+1(s_1))) = n!/2$. Since $+1(s_2)$ also has dimension $n!/2$, we conclude $P_2(+1(s_1)) = +1(s_2)$.

By induction: after P_k (for $k = 2, \ldots, n-1$), the image is all of $+1(s_k)$, each of dimension $n!/2$. At the final step, $W_1 = +1(s_{n-1})$. □

4 Cascade Surjectivity

Lemma 6 (Cascade Surjectivity)**.** *If $|k-j| > 1$ (so s_k and s_j commute), then P_k maps $+1(s_{k-1}) \cap +1(s_j)$ bijectively onto $+1(s_k) \cap +1(s_j)$.*

Proof. Since s_k commutes with s_j, P_k preserves $+1(s_j)$: if $s_j v = v$, then $s_j(P_k v) = P_k(s_j v) = P_k v$.

By the Triple Lemma, $\ker(P_k) \cap +1(s_{k-1}) = \{0\}$, so P_k is injective on $+1(s_{k-1}) \cap +1(s_j)$. This space has dimension $n!/4$ (since s_{k-1} and s_j commute, both conditions independently halve). The target $+1(s_k) \cap +1(s_j)$ also has dimension $n!/4$ (since $|k-j| > 1$ implies s_k and s_j commute). By injectivity plus equal dimensions, the map is a bijection. □

Remark 7. More generally, if $S \subseteq \{1, \ldots, n-1\}$ is a set of indices with $|k-j| > 1$ for all $j \in S$ and $|k-j| > 1$, $|(k-1)-j| > 1$ for all $j \in S$, then P_k maps $+1(s_{k-1}) \cap \bigcap_{j \in S} +1(s_j)$ bijectively onto $+1(s_k) \cap \bigcap_{j \in S} +1(s_j)$.

5 Trace Preservation

Lemma 8 (Trace Preservation)**.** *Let $V \subseteq \mathbb{C}[\mathfrak{S}_n]$ be a subspace that is s_j-invariant (i.e., $s_j(V) = V$). If P_k commutes with s_j (equivalently, $|k-j| > 1$) and P_k is injective on V, then $P_k(V)$ is s_j-invariant and*

$$\mathrm{tr}(s_j|_{P_k(V)}) = \mathrm{tr}(s_j|_V).$$

Proof. Since P_k commutes with s_j, for $v \in V$: $s_j(P_k v) = P_k(s_j v) \in P_k(V)$, so $P_k(V)$ is s_j-invariant.

Since V is s_j-invariant, it decomposes as $V = V^+ \oplus V^-$ where $V^\pm = \pm 1(s_j) \cap V$. Then $P_k(V) = P_k(V^+) \oplus P_k(V^-)$ (by injectivity of P_k on V and the fact that P_k preserves $\pm 1(s_j)$). Since P_k is injective, $\dim(P_k(V^\pm)) = \dim(V^\pm)$, so

$$\mathrm{tr}(s_j|_{P_k(V)}) = \dim(P_k(V^+)) - \dim(P_k(V^-)) = \dim(V^+) - \dim(V^-) = \mathrm{tr}(s_j|_V).$$ □

6 The halving mechanism

6.1 Regularity of traces

Lemma 9. *In the left regular representation of $\mathfrak{S}_n$, for any non-identity element $g \in \mathfrak{S}_n$: $\mathrm{tr}(g) = 0$.*

Proof. g acts by $e_w \mapsto e_{gw}$. The trace is the number of fixed points: $|\{w : gw = w\}| = 0$ since $g \neq e$. □

Corollary 10. *For any set of pairwise commuting transpositions $\{s_{j_1}, \ldots, s_{j_r}\}$ with $|j_a - j_b| > 1$ for all $a \neq b$, and any s_i with $|i - j_a| > 1$ for all a:*

$$\mathrm{tr}\Big(s_i\big|_{\bigcap_{a=1}^r +1(s_{j_a})}\Big) = 0.$$

Proof. The projector onto $\bigcap_a +1(s_{j_a})$ is $\prod_a P_{j_a}$ (commuting idempotents). The trace is

$$\mathrm{tr}\left(s_i \prod_a P_{j_a}\right) = 2^{-r} \sum_{T \subseteq \{1,\ldots,r\}} \mathrm{tr}\left(s_i \prod_{a \in T} s_{j_a}\right).$$

Each term $s_i \prod_{a \in T} s_{j_a}$ is a product of disjoint transpositions including s_i, hence non-identity. By Lemma **??**, every term vanishes. □

6.2 The first halving

Theorem 11. *For $n \geq 4$: $W_1 = +1(s_{n-1})$ is s_1-invariant, with $\mathrm{tr}(s_1|_{W_1}) = 0$, so P_1 halves the image at the start of pass 2:*

$$\dim(P_1(W_1)) = \dim(W_1)/2 = n!/4.$$

Proof. Since $|1-(n-1)| = n-2 > 1$ for $n \geq 4$, s_1 commutes with s_{n-1}, so $+1(s_{n-1})$ is s_1-invariant. By Corollary **??** with $r = 1$ and $j_1 = n-1$, $\mathrm{tr}(s_1|_{+1(s_{n-1})}) = 0$. Hence the $+1$ and -1 eigenspaces of s_1 within $+1(s_{n-1})$ have equal dimension $n!/4$. Applying P_1 kills the -1 eigenspace, giving $\dim(P_1(W_1)) = n!/4$. □

6.3 Image after pass 2

Theorem 12. *For $n \geq 4$, the image after pass 2 is*

$$W_2 = P_{n-2}(+1(s_{n-3}) \cap +1(s_{n-1})), \qquad \dim(W_2) = n!/4.$$

Moreover, $W_2 \subseteq +1(s_{n-2})$.

Proof. Pass 2 applies $P_1, P_2, \ldots, P_{n-2}$ to $W_1 = +1(s_{n-1})$.

After P_1: the image is $+1(s_1) \cap +1(s_{n-1})$, dimension $n!/4$ (by Theorem **??**).

For $k = 2, 3, \ldots, n-3$: each s_k commutes with s_{n-1} (since $k \leq n-3$ implies $|k-(n-1)| \geq 2$). By Lemma **??**, P_k maps $+1(s_{k-1}) \cap +1(s_{n-1})$ bijectively to $+1(s_k) \cap +1(s_{n-1})$.

After P_{n-3}: the image is $+1(s_{n-3}) \cap +1(s_{n-1})$, dimension $n!/4$.

Finally, P_{n-2} is injective on this space (since $+1(s_{n-3}) \cap +1(s_{n-1}) \subseteq +1(s_{n-3})$ and $\ker(P_{n-2}) \cap +1(s_{n-3}) = \{0\}$ by the Triple Lemma). So $W_2 = P_{n-2}(+1(s_{n-3}) \cap +1(s_{n-1}))$ has dimension $n!/4$, and $W_2 \subseteq +1(s_{n-2})$. □

6.4 The second halving

Theorem 13. *For $n \geq 6$: W_2 is s_1-invariant with $\mathrm{tr}(s_1|_{W_2}) = 0$, so P_1 halves the image at the start of pass 3:*

$$\dim(P_1(W_2)) = n!/8.$$

Proof. The inner space $+1(s_{n-3}) \cap +1(s_{n-1})$ is s_1-invariant (since $|1-(n-3)| \geq 3$ and $|1-(n-1)| \geq 5$ for $n \geq 6$), and $\mathrm{tr}(s_1|_{+1(s_{n-3})\cap+1(s_{n-1})}) = 0$ by Corollary **??**.

The outer projection P_{n-2} commutes with s_1 (since $|1-(n-2)| \geq 4$ for $n \geq 6$) and is injective on the inner space (by the proof of Theorem **??**). By the Trace Preservation Lemma **??**:

$$\mathrm{tr}(s_1|_{W_2}) = \mathrm{tr}(s_1|_{+1(s_{n-3})\cap+1(s_{n-1})}) = 0.$$

So W_2 has $n!/8$ each of $+1$ and -1 eigenvectors for s_1, and P_1 halves. □

6.5 General halving pattern (verified)

Conjecture 14 (Halving Lemma)**.** *For $j = 1, \ldots, \lfloor n/2 \rfloor$: the P_1 step at the start of pass j halves the image dimension. All subsequent P_1 steps (passes $\lfloor n/2 \rfloor + 1, \ldots, n-1$) are injective.*

The above theorems prove this for the first two halvings ($n \geq 4$ and $n \geq 6$ respectively). The pattern continues:

Remark 15 (Structure of later halvings)*.* Computational analysis (verified $n \leq 8$) reveals:

Image after pass j: $W_j \subseteq +1(s_{n-j})$, with $\dim(W_j) = n!/2^{\min(j,\lfloor n/2 \rfloor)}$.

V_j **formula:** Before halving $j+2$ (for $j = 0, 1, \ldots, \lfloor n/2 \rfloor - 2$), the image has the form

$$W_{j+1} = P_{n-j-1}(+1(s_{n-j-2}) \cap +1(s_{n-j})) \cap (n!/2^{j+1}\text{-dim subspace}).$$

More precisely, W_{j+1} is obtained by applying P_{n-j-1} to the cascade output: the cascade $P_1, P_2, \ldots, P_{n-j-2}$ maps $+1(s_1) \cap W_j$ (after halving) through successive eigenspace intersections, reaching $+1(s_{n-j-2}) \cap +1(s_{n-j})$ before the final step P_{n-j-1}.

s_1**-invariance:** W_{j+1} is s_1-invariant with $\mathrm{tr}(s_1|_{W_{j+1}}) = 0$, for $j + 1 \leq \lfloor n/2 \rfloor - 1$. This follows from:

1. The "inner" intersection $+1(s_{n-j-2}) \cap +1(s_{n-j})$ is s_1-invariant with vanishing s_1-trace (by Corollary **??**, provided $n - j - 2 \geq 3$, i.e., $j \leq n - 5$).
2. The "outer" projection P_{n-j-1} commutes with s_1 (provided $n - j - 1 \geq 3$, i.e., $j \leq n - 4$).
3. P_{n-j-1} is injective on the inner space (by the Triple Lemma).
4. Trace Preservation (Lemma **??**) propagates the vanishing trace through P_{n-j-1}.

For $j + 1 \leq \lfloor n/2 \rfloor - 1$, the conditions $j \leq n - 5$ and $j \leq n - 4$ are satisfied (since $\lfloor n/2 \rfloor - 2 \leq n - 5$ for $n \geq 6$).

Verification. The halving pattern is confirmed for $n = 2, 3, \ldots, 8$:

n	$\lfloor n/2 \rfloor$	Rank after each P_1 step
2	1	$2 \to 1$
3	1	$6 \to 3$
4	2	$24 \to 12 \to 6$
5	2	$120 \to 60 \to 30$
6	3	$720 \to 360 \to 180 \to 90$
7	3	$5040 \to 2520 \to 1260 \to 630$
8	4	$40320 \to 20160 \to 10080 \to 5040 \to 2520$

Additionally, the cascade surjectivity formula $W_2 = P_{n-2}(+1(s_{n-3}) \cap +1(s_{n-1}))$ is verified for $n = 4, 5, 6$, with s_1-invariance and $\mathrm{tr}(s_1) = 0$ confirmed.

7 The rank formula

Theorem 16 (Rank Formula)**.** *Assuming Conjecture 6 (proved for the first two halvings, verified for $n \leq 8$):*

$$\mathrm{rank}(\Pi) = \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Proof. By Corollary 5, $\mathrm{rank}(\Pi) = n!/2^d$ where d is the number of halving P_1 steps. By Conjecture 6, $d = \lfloor n/2 \rfloor$. □

Remark 17. The count $n!/2^{\lfloor n/2 \rfloor}$ equals the number of *descent-paired permutations* of $[n]$: those w with $w(2i-1) > w(2i)$ for all $i = 1, \ldots, \lfloor n/2 \rfloor$ (OEIS A090932, Callan's characterization). Computational verification ($n \leq 5$) confirms that $\{\Pi \cdot e_w : w \text{ descent-paired}\}$ is in fact a *basis* for $\mathrm{im}(\Pi)$.

8 Eigenspace cascade tracking

The cascade pattern through all steps reveals the following structure (verified for $n \leq 6$):

Proposition 18 (Eigenspace tracking)**.** *After step P_k within pass j (i.e., after applying $P_1, \ldots, P_k$ to W_{j-1}), the image is contained in $+1(s_k)$. Moreover, if $k \leq n-j-1$, the image is also contained in $+1(s_{n-j+1})$ (the eigenspace condition "carried" from the previous pass). At the last step P_{n-j}, the carried condition $+1(s_{n-j+1})$ is lost (since s_{n-j} is adjacent to s_{n-j+1}), and the image enters $+1(s_{n-j})$.*

For $\mathfrak{S}_6$, the full tracking is:

Step	Projection	Image fixed by
1	P_1	s_1
2	P_2	s_2
3	P_3	s_3
4	P_4	s_4
5	P_5	s_5 (end of pass 1)
6	P_1	s_1, s_5 (**halving**)
7	P_2	s_2, s_5
8	P_3	s_3, s_5
9	P_4	s_4 (s_5 lost, end of pass 2)
10	P_1	s_1, s_4 (**halving**)
11	P_2	s_2, s_4
12	P_3	s_3 (s_4 lost, end of pass 3)
13	P_1	s_1 (**halving**)
14	P_2	s_2 (end of pass 4)
15	P_1	s_1 (no halving, end of pass 5)

9 Remaining gaps

1. **General halving (passes ≥ 4).** The proof of the first two halvings uses the explicit formula $W_2 = P_{n-2}(+1(s_{n-3}) \cap +1(s_{n-1}))$ to establish s_1-invariance and vanishing trace via Trace

Preservation. Extending this to the third and higher halvings requires showing that the cascade $P_2, P_3, \ldots$ (which includes P_2 that does *not* commute with s_1) produces an s_1-invariant output. The difficulty: P_2 temporarily breaks s_1-invariance, and while $P_3, P_4, \ldots$ (which commute with s_1) should "restore" it, a clean algebraic proof of this restoration is lacking.

2. **Halving cessation.** After $\lfloor n/2 \rfloor$ halvings, the image has no $-1(s_1)$ eigenvectors. This is verified computationally but not proved. The mechanism appears related to the "carried" eigenspace condition reaching indices too close to 1 for s_1-commutativity.

3. **Image basis.** The conjecture that descent-paired permutations index an image basis is verified for $n \leq 5$.

10 Proof summary and status

Result	**Status**
Triple Lemma (2)	Proved
Within-Pass Injectivity (4)	Proved
First Block Surjectivity (**??**)	Proved
Cascade Surjectivity (**??**)	Proved
Trace Preservation (**??**)	Proved
Regularity of traces (**??**, **??**)	Proved
First halving (**??**)	Proved ($n \geq 4$)
Image after pass 2 (**??**)	Proved ($n \geq 4$)
Second halving (**??**)	Proved ($n \geq 6$)
General halving (6)	Verified $n \leq 8$
Rank formula (8)	Conditional on 6
Image basis	Verified $n \leq 5$

References

[1] P. Galashin, *Symmetries of stochastic colored vertex models*, Ann. Probab. **49** (2021), no. 5, 2175–2219.

[2] OEIS Foundation, *A090932:* $n!/2^{\lfloor n/2 \rfloor}$, `https://oeis.org/A090932`.